1. REQUIREMENTS ELICITATION

FUNCTIONAL REQUIREMENTS

No.	Requirement
1	The system shall allow a customer to register either as an individual, by providing first name, surname, address, identification details, and contact details, or as a company/organisation, by providing company name, company address, registration details, and contact details.
2	The system shall validate customer information and prevent registration when required fields are missing or invalid.
3	The system shall allow a registered user, either a customer or authorised bank staff member, to sign in using a username and password. The system shall reject invalid login attempts and display an appropriate message.
4	The system shall allow a logged-in user to log out securely.
5	The system shall lock a user account after three consecutive failed login attempts.
6	The system shall restrict functions according to the user's role, such as customer or authorised bank staff member.
7	The system shall allow an authenticated customer to view their profile and list all accounts held under their customer profile, including account number, account type, and current balance.
8	The system shall allow authorised bank staff to search for a customer by customer ID or surname.
9	The system shall allow a registered customer to open multiple savings, investment, and cheque accounts, including more than one account of the same type and all three account types together, applying the rules specific to each type. Each account shall be linked to exactly one existing customer, and the system shall refuse to create an account without an associated customer.

10	The system shall generate a unique account number for every new account and record the branch and date opened.
11	The system shall require an initial deposit of at least BWP 500.00 to open an investment account.
12	The system shall permit a cheque account to be opened only for an employed individual customer and shall record the employer name and employer address.
13	The system shall allow a customer to deposit funds into any account they hold, subject to validation of the deposit amount.
14	The system shall allow withdrawals from investment and cheque accounts only. The system shall reject a withdrawal request from a savings account, leave the balance unchanged, and display a clear message.
15	The system shall reject a withdrawal request where the amount exceeds the available account balance, leave the balance unchanged, and display an appropriate message.
16	The system shall reject a deposit or withdrawal amount that is zero, negative, or not a valid monetary value.
17	The system shall update an account balance after every successful deposit, withdrawal, or interest credit.
18	The system shall record every deposit, withdrawal, account opening, and interest credit as a transaction history event with a unique transaction ID, transaction type, amount where applicable, date and time, affected account, and resulting balance.
19	The system shall allow a customer to view the transaction history for each account they hold.
20	The system shall display clear confirmation or error messages for account-opening and transaction requests.
21	The system shall calculate and credit interest to eligible accounts once every month.

22	The system shall pay monthly interest at 0.05% of the balance for each savings account.
23	The system shall pay monthly interest at 5% of the balance for each investment account.
24	The system shall not pay interest on cheque accounts.
25	The system shall allow authorised bank staff to assist with customer or account administration according to their assigned access permissions.

NON-FUNCTIONAL REQUIREMENTS

Quality attribute	Requirement
Security	Authentication: The system shall authenticate users before providing access to customer or account information. Authorisation: A customer shall only be able to view and operate on accounts linked to their own profile. Staff functions shall not be available to customers and shall be restricted according to assigned permissions. Data Protection: Customer and account data shall be protected from unauthorised access, and sensitive data shall not appear in error messages or logs. Credential protection: Passwords shall not be stored as readable plain text. The implementation should use an appropriate password-hashing method. Session Handling: A session shall end automatically after 10 minutes of inactivity.

Privacy	The system shall collect and display only customer information needed for its stated functions.
Data integrity and Accuracy	Account balances and transaction records shall remain consistent. A rejected or failed transaction shall not partially update the account. A transaction shall either complete in full or not at all. Monetary values shall be stored and calculated with exact decimal precision and shown to two decimal places in Botswana Pula (BWP).
Performance	Actions such as login, deposit, withdrawal, balance enquiry, and transaction-history viewing should complete within two seconds under normal conditions.
Usability	Screens shall use clear labels, and error messages shall say what went wrong and how to fix it.
Reliability	The application shall handle invalid input and database errors without crashing. If the system becomes temporarily unavailable or crashes, it shall recover without losing committed transactions and shall not report an uncompleted transaction as successful.
Availability	The system shall be accessible to customers 24 hours a day, 7 days a week, except during planned maintenance or unexpected service interruptions. It shall display a clear message when a service or transaction cannot be completed.
Maintainability	The design and code shall follow object-oriented principles (abstraction, inheritance, polymorphism, encapsulation and interfaces), with separate user-interface, controller, domain, and data-access responsibilities so that each part can be changed and tested independently without affecting the others.
Portability	The system shall run on any computer with a suitable Java runtime and shall also run in GitHub Codespaces.
Scalability	The system should be designed to support growth in the number of customers, accounts, and transaction records without requiring major changes to its core structure.

Auditability	All financial operations and interest credits shall be traceable and recorded with enough detail to support review of account activity.
Extensibility	The system shall be designed so that new account types, interest-calculation rules, and transaction types can be added with minimal changes to existing classes.

INTERVIEW RECORD

Interview purpose: To identify the Banking System's core functions, business rules, and quality attributes.

Interviewee: Kentsenao Baseki

Interviewer: Neoyame Dibotelo

Date:

Method: Mock interview

Question	Response
1. Who will use the system?	Customers, who may be individuals or companies will use it to access their own profiles and accounts. Bank staff may need to support account administration.
2. What information is needed to register a customer?	At minimum, the customer's first name, surname, address, and contact details should be recorded. Additional identity details can be required by bank policy. For a company, the system should record company name, company address, registration details, and contact details.

3. Can a customer hold more than one account?	Yes. One customer can hold multiple accounts, including multiple accounts of the same type and combinations of all three account types.
4. Can an account exist without a customer?	No. Every account must belong to a customer.
5. What account types must the first version of the system support?	Savings, investment, and cheque accounts.
6. What restrictions apply to savings accounts?	Customers can deposit money and receive monthly interest, but they cannot withdraw from a savings account. It pays 0.05% interest every month on the balance.
7. What restrictions apply to investment accounts?	Customers can deposit and withdraw. The account requires an initial deposit of at least BWP 500.00 and receives 5% monthly interest.
8. Who may open a cheque account?	An employed customer may open one as it is made for salaries. The system must record the employer's name and address. Deposits and withdrawals of any amount are allowed. It does not pay interest.
9. Can a cheque account go into overdraft?	No. A withdrawal larger than the balance must be refused.
10. What should happen when a new customer comes to the bank?	The teller registers the customer with personal details, then opens the account the customer wants.
11. Should users log in?	Yes. Nobody should use the system without logging in. After three wrong attempts, the user should be locked out.

12. How should deposits and withdrawals be handled?	Validate the amount and account type before changing the balance. Record successful transactions so they can be reviewed later.
13. What interest rates should be used?	0.05% monthly for savings and 5% monthly for investment accounts.
14. What if something fails during a transaction?	The balance must never be left half updated. The transaction either fully completes or it does not happen at all.
15. What should happen when a transaction is rejected?	The balance should remain unchanged, and the user should receive a clear explanation of why the request was rejected.
16. What information should transaction history show?	The account, transaction type, amount, date and time, and resulting balance would be useful.
17. What security and usability expectations matter?	Users should only access their own accounts, Credentials such as passwords should be protected, and common tasks should be straightforward to complete.
18. How accurate must the money calculations be?	There must be no rounding errors, and amounts are shown in Pula with two decimal places. The calculations must be exact.
19. How should the system be built?	It should be object-oriented, easy to extend with new account types, and run on any computer with Java.
20. What are the expectations for speed?	Everyday actions such as a deposit or balance check should respond in about two seconds.

Business Rules

- A customer may be an individual or a company/organisation.
- A customer may hold one or many accounts and may hold all three account types: Savings, Investment, and Cheque.
- Every account must belong to exactly one customer. An account cannot exist independently of its owner.
- A Savings Account accepts deposits and earns monthly interest of 0.05% of its balance. It does not permit withdrawals.
- An Investment Account accepts deposits and withdrawals, earns monthly interest of 5% of its balance, and can only be opened with an initial deposit of at least BWP 500.00.
- A Cheque Account accepts deposits and withdrawals of any amount and is normally used for salary crediting. It may only be opened for an employed individual customer, who must provide the employer's name and address.
- A Cheque Account does not earn interest.

UML DIAGRAMS

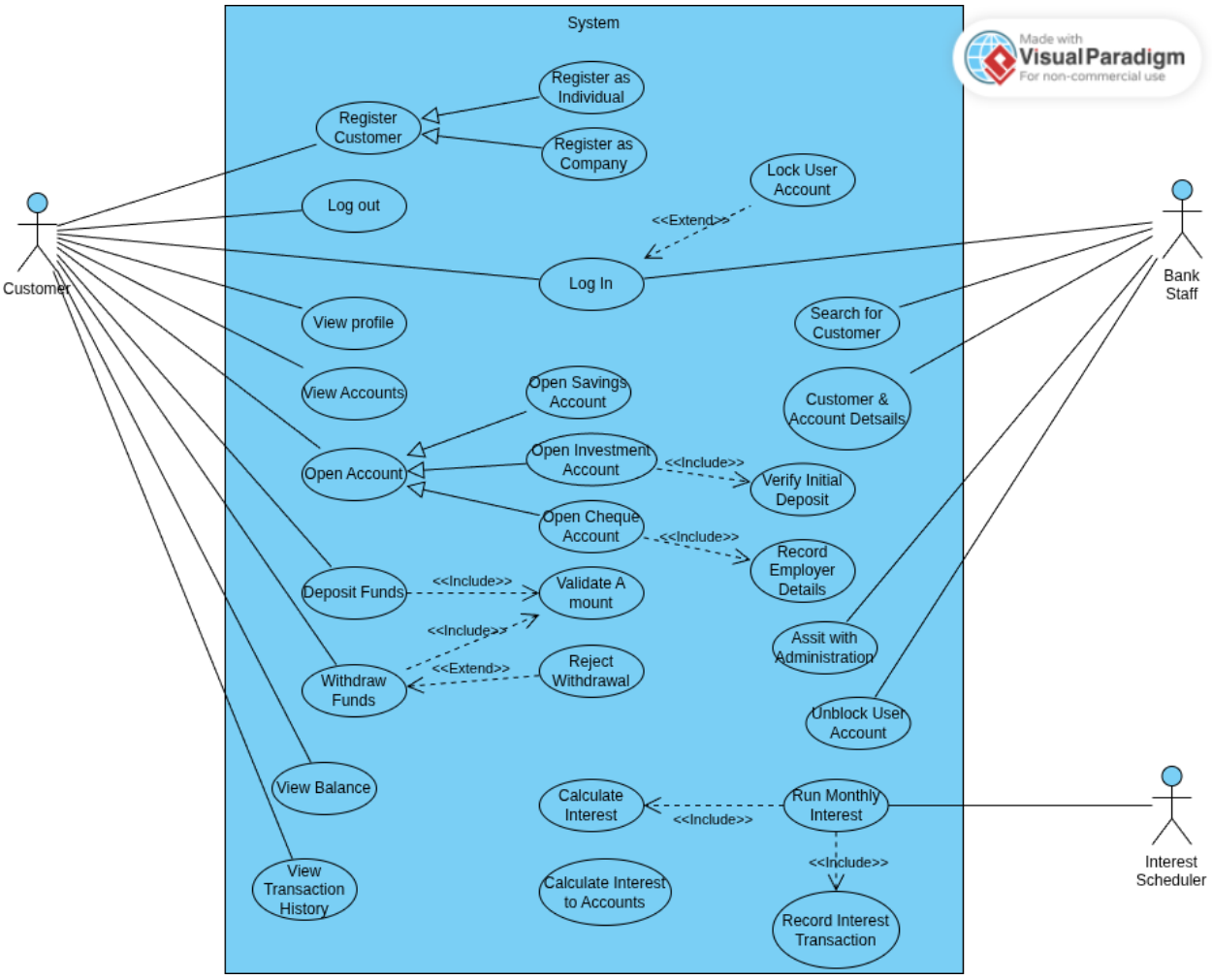
Actor	Use Cases
Customer	Register as an individual, register as a company/organization, log in, log out, view profile, view accounts, open a savings, investment, or cheque account, deposit funds, withdraw funds from an investment or cheque account only view balance, view transaction history.
Bank Staff	Log in, search for a customer, view customer and account details, assist with account administration, and unblock a blocked user account.
Interest Scheduler	Run monthly interest processing, including calculating interest, crediting interest to eligible accounts, and recording the interest transaction.

2.1 System Use Case Diagram

The **Use Case Diagram** shows how the Customer, Bank Staff, and Interest Scheduler interact with the Banking System. Customers register, manage their accounts, make permitted transactions, and view account information. Bank Staff assist with customer and account administration, while the Interest Scheduler initiates monthly interest processing.

The system boundary separates the application's functions from its external actors. <<include>> relationships show required supporting functions, such as validating credentials and recording transactions. The <<extend>> relationship shows account locking after three consecutive failed login attempts. Generalisation shows savings, investment, and cheque account opening as specialised forms of Open Account.

The diagram also captures account restrictions, including the investment opening deposit, cheque-account employment requirements, and the prohibition of savings-account withdrawals.



2.2 Class Diagram

The **Class Diagram** shows the Banking System's classes, their attributes and operations, and the relationships between them. IndividualCustomer and CompanyCustomer inherit from Customer, while SavingsAccount, InvestmentAccount, and ChequeAccount inherit from the abstract Account class. Each account belongs to one customer and maintains its transaction history.

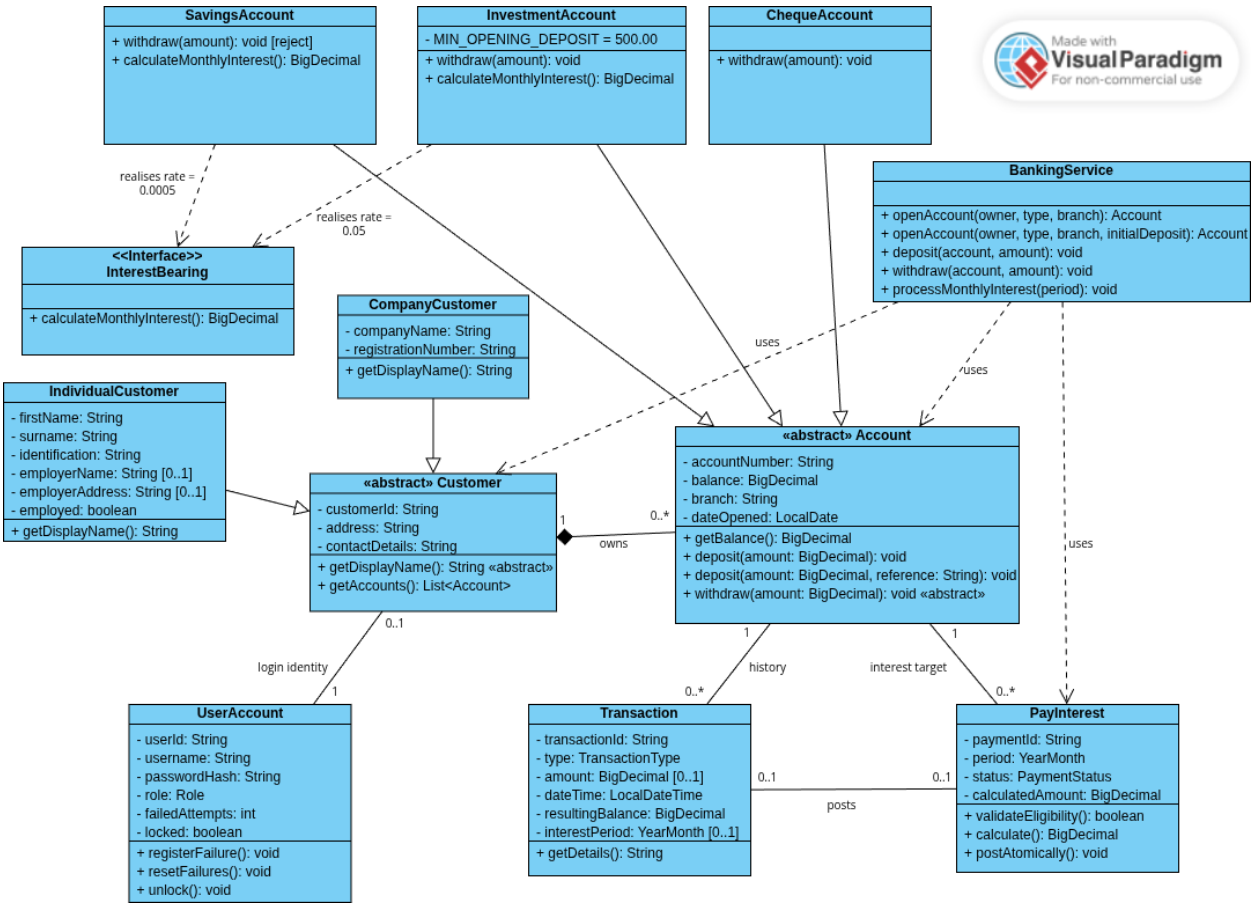
The design demonstrates abstraction through the abstract classes, encapsulation through private attributes, and inheritance through the customer and account subclasses. SavingsAccount and InvestmentAccount implement the InterestBearing interface, providing their own interest calculations. Overriding supports account-specific

behaviour, while overloaded deposit and account-opening operations accept different input combinations.

UserAccount manages login details and account locking, BankingService coordinates banking operations, and PayInterest manages monthly interest payments.

Class / Interface	Responsibility
Customer	Abstract superclass that stores shared customer information, including customer ID, address, and contact details. Associates a customer with their accounts.
IndividualCustomer	Represents an individual customer. Stores personal identification and employment details and provides the individual's display name.
CompanyCustomer	Represents a company or organisation. Stores the company name and registration number and provides the company's display name.
UserAccount	Stores login credentials, user role, failed-login count, and lock status. Supports recording failed attempts, resetting the count, and unlocking access.
Account	Abstract superclass that stores the account number, balance, branch, and opening date. Provides shared deposit and balance operations and declares account-specific withdrawal behaviour.
SavingsAccount	Allows deposits, prohibits withdrawals, and calculates monthly interest at 0.05% of the balance.
InvestmentAccount	Requires an opening deposit of at least BWP 500.00, permits deposits and withdrawals, and calculates monthly interest at 5% of the balance.
ChequeAccount	Allows deposits and withdrawals without earning interest. Its owner must be an employed individual who supplies employer details.

InterestBearing	Interface declaring the monthly interest-calculation operation implemented by SavingsAccount and InvestmentAccount.
Transaction	Records account-history events, including account openings, deposits, withdrawals, and interest credits. Stores the event's ID, type, amount where applicable, date and time, and resulting balance.
BankingService	Coordinates account opening, deposits, withdrawals, and monthly interest processing. Ensures the relevant account rules and recording operations are applied.
PayInterest	Represents an interest payment for one account and month. Tracks its processing state, validates eligibility, calculates interest, and coordinates posting the balance update and interest record together.

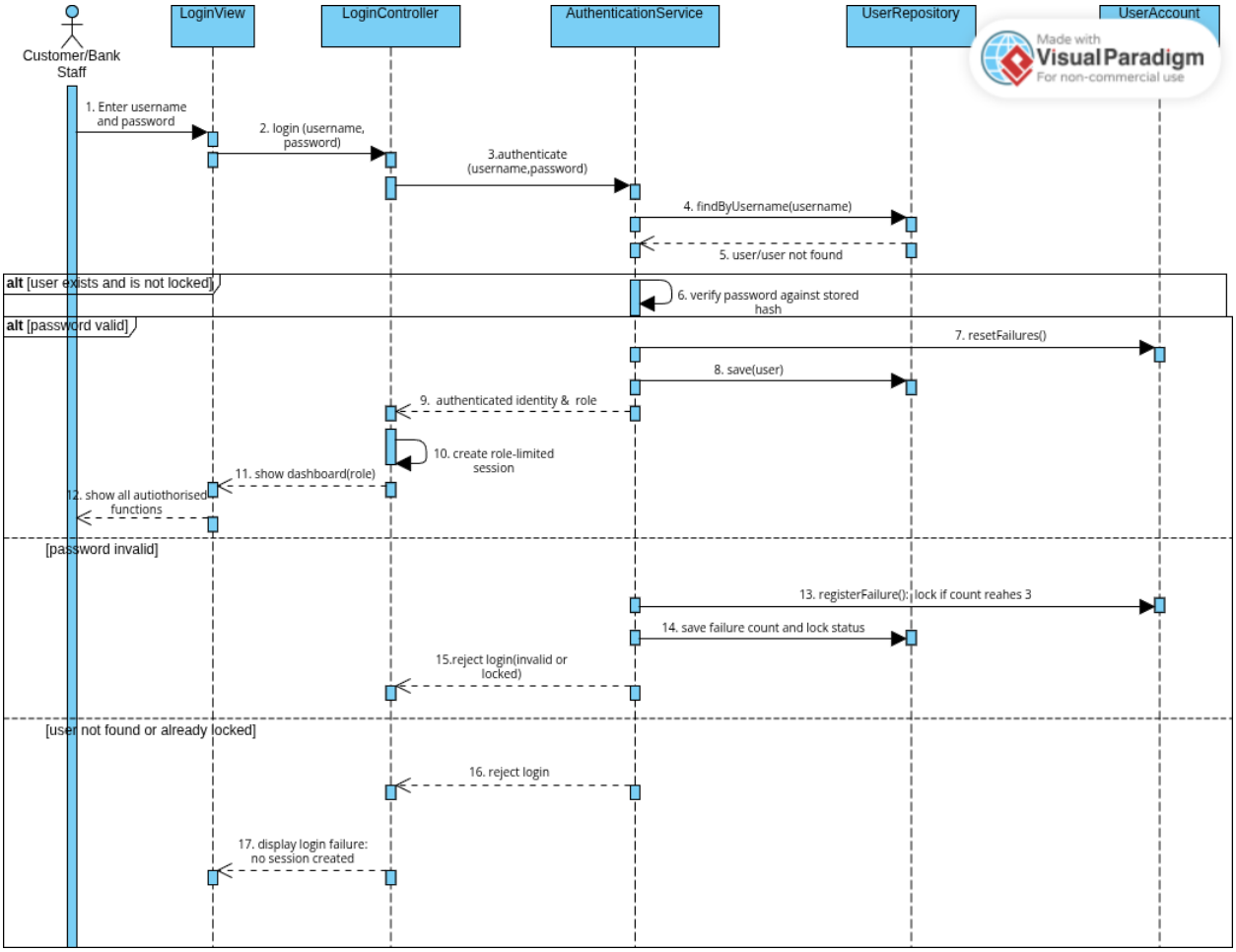


3.1(a) Login Sequence Diagram

The **Login Sequence Diagram** shows the interactions between the user, LoginView, LoginController, AuthenticationService, UserRepository, and UserAccount during login.

The user enters a username and password, which are passed to the authentication service for verification. If the credentials are valid and the account is unlocked, the system resets the failed-attempt count, creates a role-restricted session, and displays the appropriate dashboard.

Alternative paths show invalid credentials, an unknown username, or a locked account. Failed attempts are recorded, and the account is locked after three consecutive failures. Rejected login attempts display an error message without creating a session.

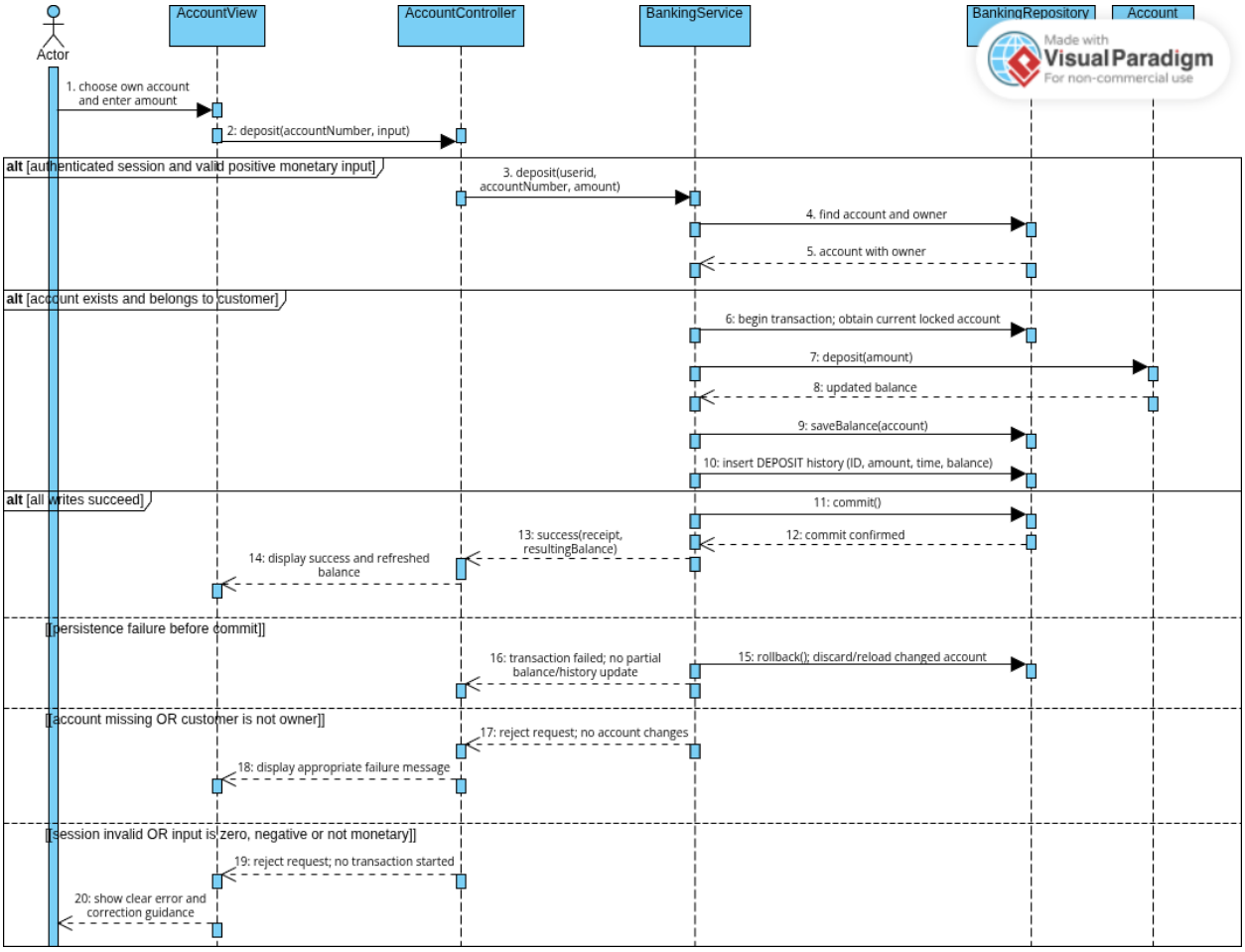


3.1(b) Deposit Funds Sequence Diagram

The **Deposit Funds Sequence Diagram** shows how a customer's deposit request passes through AccountView, AccountController, BankingService, the repository, and the Account.

The system checks that the user is authenticated, the account belongs to the customer, and the deposit amount is valid. If these checks pass, it updates the account balance and records the deposit. Both changes are committed together; if saving fails, the operation is rolled back so the balance and transaction history remain consistent.

If the account cannot be found, does not belong to the customer, or the amount is invalid, the deposit is rejected and the system displays an error message.



3.2 PayInterest State Diagram

The **PayInterest State Diagram** shows the lifecycle of a monthly interest payment for one account. Processing begins in Pending and moves through Validating, Calculating, and Posting before reaching Completed.

During validation, cheque accounts and accounts already credited for that month move to Skipped. Eligible savings and investment accounts proceed to interest calculation. During Posting, the balance update and interest transaction are committed together.

An error moves the payment to Failed. A retry returns it to validation, where existing payment records are checked to prevent duplicate interest credits.

